fast iterative mod 3

```
while (true)
{
  if(rax ≤ 5) break;
  rax = (rax ∧ 3) + rax >> 2
}
rax = rax % 3;
```

$/2^2$ [0x23]

$\%4$

while (true)
{
if (rax ≤ 5) break
rax = (rax ∧ 3) + rax >> 2
}

rax / 2^2

rdx

if rax ≤ 5

rax ∈ {3,4,5} ⇒ CF = 0

↓ cmc

CF = 1

CF = 1 ⇐ rax < 3

↓

CF = 0 rdx = rdx − ((rdx + 0) ∧ 3))

rdx = rdx − ((0 + 1 ∧ 3))

i.e. rdx = rdx − 1

rax = rax − rdx

5 3 rdx ∈ 3 ⇒ rdx ∈ {0,1,2,3}

mod 5?

```
.loop:
        cmp     rax, 5
        jbe     .exit_loop
        mov     rdx, rax
        shr     rdx, 2
        and     rax, 3
        add     rax, rdx
        jmp     .loop
.exit_loop:
```

rax ≤ 5 ⇒ exit loop

ELSE

rdx := rax > 5

rdx = rax / 2^2

rax = rax ∧ 0011

rax = rax + rdx

i.e. loop:

rax = (rax ∧ 0011) + rax / 2^2

rdx = rdx − (rdx + CF)

rdx = rdx ∧ 3

rax = rax − rdx

= rax − (rdx ∧ 3)

= rdx − ((rdx − rdx + CF) ∧ 3))

```
        cmp     rax, 3
        cmc     "complement carry flag"
        sbb     rdx, rdx
        and     rdx, 3
        sub     rax, rdx
```

ZF := 1 ⇔ rax = 3

SF := 1 ⇔ rax < 3

CF := 1

mod 7

i.e. ¬CF

0x23

.loop:

```
cmp   rax,5
jbe   .exit_loop
mov   rdx,rax
shr   rdx,2
and   rax,3
add   rax,rdx
jmp   .loop
```

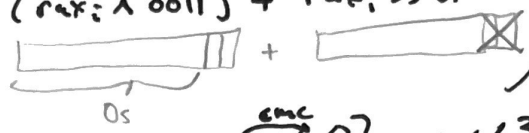
$rax_i \le 5 \Rightarrow$ EXIT loop

$rdx = rax_i$

$rdx = rax_i >> 2$

$rax = rax_i \wedge 3 = rax_i \wedge 0011$

$rax = (rax_i \wedge 0011) + rax_i >> 2$



$x_{4i+1} = (x_i \wedge 0011) + (x_i >> 2)$

.exit_loop:

```
cmp   rax,3
cmc
sbb   rdx,rdx
and   rdx,3
sub   rax,rdx
```

$rax < 3 \Rightarrow CF = 1 \xrightarrow{cmc} 0$

$rax \ge 3 \Rightarrow CF = 0 \rightarrow 1$

$rax < 3 \Rightarrow CF = 0$

$rax \ge 3 \Rightarrow CF = 1$

$rdx = rdx - (rdx + CF)$ $\xrightarrow{rax<3}$ 0 ; $\xrightarrow{rax \ge 3}$ $xff\ldots ff$

$rdx = 0$ $(rax < 3)$

$= 3$ $(rax \ge 3)$

$rax = rax - rdx$

$\Leftrightarrow$

$rax < 3$:

$= \{0,1,2\} - 0 = rax \in \{0,1,2\}$

$5 \ge rax \ge 3$: ($\{3,4,5\}$)

$= \{3,4,5\} - 3 = \{0,1,2\}$

ie mod 3

ie preserves the value % 3

ie super-fast % 3

op:

cmp rax, 5 — (rax − 5) accumulator

- > 5: SF, CF == 0
- $= 5$: ZF = 1
- < 5: CF = 1

jbe .exit_loop — "jump if below OR equal" ⟺ CF OR ZF = 1

mov rdx, rax — rax → rdx ie rdx := rax (accumulator)

shr rdx, 2 — "..." ie rdx / 2^2

and rdx, 3 — rdx = {0, 1, 2, 3}

add rax, rdx — rax = rax + rdx ie rax = {0, 1, 2, 3}

jmp .loop — unconditional

{TRUE / FALSE ...}

.exit_loop:

cmp rax, 3

cmc — "complement carry flag" ie CF=1 → CF=0; CF=0 → CF=1

sbb rdx, rdx — rdx = 0 (CF=0); 0xF...F ~ 11... ~ −1 (CF=1)

and rdx, 3 — rdx := {0, 1, 2, 3}

sub rax, rdx — rax := rax − rdx = rax − {0, 1, 2, 3}

?